

PBM paper outline

Ecosystem structure

- Functional groups (each an independent network?; reproduction and growth rate)
 - Age groups (unit of the food web; specific biomass, energy coefficients)
 - § Grid-cell population (integer aggregate of pop units, rounded to 0 if <1)
 - Population unit (a pack, school, or individual; unit of biomass)

Action space (Grid-cell population):

- Deterministic, pre-defined:
 - Age group transition
 - Reproduction
 - Mortality below energy lower threshold
 - Potentially seasonal migration
- Policy network (FG specific, each with an energy cost):
 - Move
 - Eat (with a success rate)
 - Hide/rest (with effects on prob of being hunted)

Biomass $\rightarrow$ n. population units $\times$ energy level

Functional groups

1. Phytoplankton
2. Zooplankton
3. Benthic community
4. Pelagic fish
5. Codfish
6. Porpoises
7. Seals
8. Seabirds

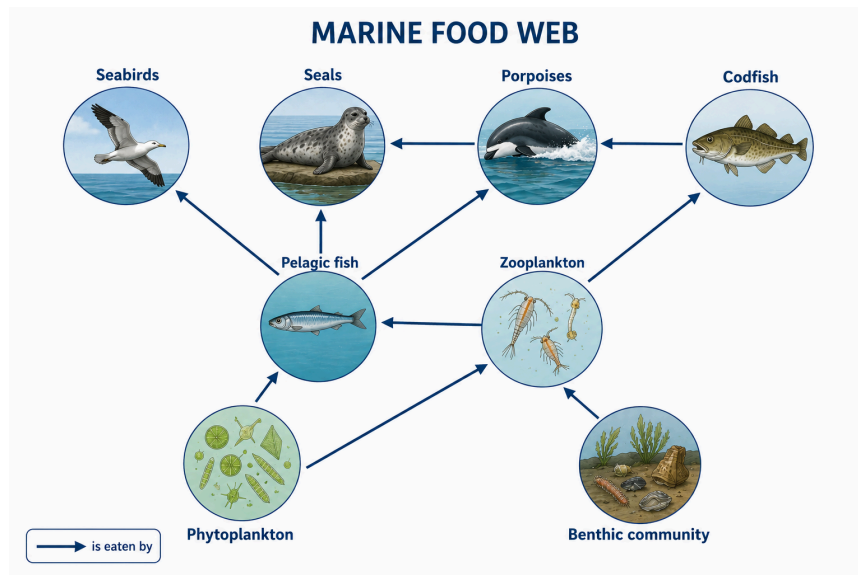
Consider adding detritus as an FG to make the energy flow more complete. Detritus is eaten by the benthic community. Add age groups.

Age groups

FG 4-8 are divided into age groups. Juvenile and Adult.

Thus, we get a total of 13 FGs, 10 decision-making and 3 non-decision-making FGs.

Food web (illustration)



Yearly reproduction

Example: Atlantic cod can live up to 25 years. Reaches sexual maturity after about 3 years.

- Birth: $x\%$ of all Adult cod biomass is turned to Juvenile cod biomass once per year
- Growth: $\frac{1}{3}$ of all Juvenile cod biomass changes to Adult cod biomass
- Mortality is modeled continuously, not just once per year.

Training

The policy networks of the 10 decision-making FGs will be trained separately.

The training goal of FG X is essentially to maximize the total energy delta of X within H steps and across several worlds

Total energy = structural energy + reserve energy

FG transitions are not part of the training.

Reserve

Reserve (unavailable for predation): 5% of max of each FG is unavailable for predation. But they may go down to 0 for other reasons including energy loss (starvation).

Three ways of defining the max biomass of FG X in a cell:

1. Hard-coded max (CC). This is used for producers, but for consumers we want it to be emergent
2. Spawned max: max initial spawning (depending on suitability)
3. Observed (floating) max

Hunting success

Two alternatives:

1. The hunting success is 100% above and 0 below the reserve level.
2. The hunting success goes down slowly towards the reserve level as the prey gets more sparse.

For example:

$$A_{XY}(c) = B_X^{imp}(c) \pi_{X,eatY}(c) I_{XY} h_X(c)$$

$$B_Y^{avail}(c) = \max(0, B_Y^{imp}(c) - B_Y^{prot}(c))$$

$$S_Y(c) = \frac{B_Y^{avail}(c)}{B_Y^{avail}(c) + T_Y(c)}$$

where

$$T_Y(c) = \sum_Z A_{ZY}(c)$$

Then:

$$C_{XY}(c) = A_{XY}(c) S_Y(c)$$

This combines both ideas:

1. **Refuge:** some biomass is completely unavailable.
2. **Hunting success:** even available prey becomes harder to catch when scarce.

A_{XY} = attempt

T_Y = total attempt

S_Y = success factor

C_{XY} = actual consumption

Area

Alternatives:

- The Poseidon wind park (HaV's favorite and also interesting for Länsstyrelsen)
- The Mareld wind park (Länsstyrelsen's favorite)
- Poseidon + Mareld wind park

Scenarios

- Baseline (trawling at current level)
- Construction phase (no trawling + noise)
- Operational phase (no trawling + rotor action)

- Conservation (no fishing + no wind farm)

Results

- Population dynamics
 - Biomass development curves
 - Biomass distribution maps (images + videos)
- Population behavior ('IQ tests')
 - Basic behavior
 - Eat food
 - Approach food
 - Escape predator
 - Action distribution
 - Eat
 - Rest
 - Move

Extensions

Memory

Taxis